

Jinuk Moon

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Seoul National University

PROFILE

PhD candidate developing [machine learning interatomic potentials \(MLIPs\)](#) for accelerating catalyst [discovery](#). Developed [CatBench](#), a benchmarking framework for reliable evaluation of MLIPs in heterogeneous catalysis, and [oh-my-mlip](#), an LLM agent-driven framework for MLIP installation, benchmarking, fine-tuning, and distillation. Research focuses on [knowledge distillation of foundation MLIPs](#) and [multiscale simulation combining machine learning with first-principles calculations](#) to discover efficient catalytic materials for clean energy applications.

EDUCATION

- **Seoul National University** 2023 – Present
Ph.D. Materials Science and Engineering Seoul, South Korea
 - Machine learning-accelerated computational catalysis for electrocatalyst discovery and design
- **Pohang University of Science and Technology (POSTECH)** 2019 – 2022
B.S. Chemical Engineering Pohang, South Korea
 - Thesis: Study of Metal-Nitrogen-doped carbon catalysts for Chlorine Evolution Reaction combining DFT calculation with Machine Learning
 - GPA: 3.98/4.3 (Summa Cum Laude)

OPEN SOURCE & CODE

- **CatBench - Benchmark Framework for Machine Learning Interatomic Potentials in Catalysis**
Standardized framework for fair comparison of MLIPs in heterogeneous catalysis
 - pip-installable Python package | catbench.org | github.com/JinukMoon/CatBench
- **oh-my-mlip - Unified Hub for Machine Learning Interatomic Potentials**
LLM agent-driven framework for MLIP installation, execution, benchmarking, fine-tuning, and distillation
 - Agent plugin | jinukmoon.github.io/oh-my-mlip | github.com/JinukMoon/oh-my-mlip

HONORS & AWARDS

- **Presidential Science Scholarship (PhD Track; KRW 24M/yr, ~USD 17K/yr)** 2025 – Present
Ministry of Science and ICT South Korea
- **Soseon Foundation Scholarship (KRW 20M, ~USD 14K)** 2026
Soseon Foundation – Graduate research scholarship for outstanding STEM students South Korea
- **Best Oral Award** Spring 2025
The Korean Institute of Metals and Materials (KIM) South Korea
 - CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis
- **Best Oral Award** Fall 2024
The Korean Institute of Metals and Materials (KIM) South Korea
 - Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential

• **Best Oral Award**

The Korean Ceramic Society (KCerS)

Spring 2024

South Korea

- Selective Control of Oxygen Evolution Mechanisms by Doping Concentrations in RuO₂

• **Best Poster Award**

2025

The 8th International Conference on Advanced Electromaterials (ICAE 2025)

South Korea

- Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution

PUBLICATIONS (12 PUBLISHED, 2 IN REVISION, 8 FIRST-AUTHORED)

† Equal contribution, * Corresponding author

- [revision] Jun, H.[†], Kwon, D.[†], **Moon, J.[†]**, Han, S.[†], Kim, S., Yi, S. Y., Kim, Y., Kang, E., Choi, C. H.* , Han, J. W.* , Kim, W.* , Lee, J.* (2026). **Synergistic Effect of Tuning Reaction Mechanism and Interfacial Water for Efficient Acidic Water Oxidation.** *Small* (in revision).
- [revision] Jo, H.[†], Sohn, Y.[†], **Moon, J.[†]**, Han, J. W.* , Kim, W.* , Yong, K.* (2026). **Synthesis-stage Interfacial Engineering of Edge-rich β -Co(OH)₂ for Efficient Nitrate to Ammonia Electrosynthesis.** *Advanced Functional Materials* (in revision).
- [1] **Moon, J.**, Han, J. W.* (2026). **Charge-aware knowledge distillation of foundation interatomic potentials.** *AI for Accelerated Materials Design (AI4Mat) Workshop, NeurIPS 2026.*
- [2] Kim, J.[†], **Moon, J.[†]**, Cho, K.[†], Kim, J.[†], Han, M. H., Kim, S., Han, J., Choi, D., Jun, H., Jang, J. H., Oh, H.-S., Cho, S. K.* , Kim, W.* , Han, J. W.* , Lee, J.* (2026). **Size-Dependent Dynamic Surface States of Ru Clusters: Modulating Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution Reaction.** *Energy & Environmental Science*, 19, 4211.
- [3] Kim, J.[†], Yang, G. G.[†], Lee, W.[†], Nam, S., **Moon, J.**, Yang, H., Seo, J., Choi, Y., Jeong, B.* , Han, J. W.* , Kim, S. O.* , Jung, W.* (2026). **Quantitative Electrochemical Evaluation of Metal Nanocatalysts for Oxygen Exchange on Solid Oxide Cell Electrodes.** *Energy & Environmental Science*, 19, 4932.
- [4] Lim, H.[†], Choung, S.[†], **Moon, J.**, Han, J. W.* (2026). **Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration.** *npj Computational Materials*, 12, 193.
- [5] Park, K.[†], Lee, W.[†], Park, J., Jo, M., Bae, Y., Kim, M., Shin, S., **Moon, J.**, O'Hayre, R.* , Han, J. W.* , Song, S.* , Park, J. Y.* (2026). **Composite air electrodes based on BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3- δ} for reversible solid oxide cells.** *Nature Energy*, 11, 996.
- [6] Jung, H.[†], Lee, Y.[†], Cho, K.[†], Kim, T. Y., Song, J., Jung, H. S., Park, S. I., Lee, Y., **Moon, J.**, Park, W., Nam, J., Park, S., Kim, W., Han, J. W.* (2026). **Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction Under Seawater.** *Advanced Functional Materials*, 36, e16387.
- [7] Choung, S.[†], Kim, M.[†], **Moon, J.[†]**, Han, J. W.* (2025). **From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials.** *ACS Energy Letters*, 10, 6288.
- [8] **Moon, J.**, Jeon, U., Choung, S., Han, J. W.* (2025). **CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis.** *Cell Reports Physical Science*, 6, 102968.
- [9] Jun, H.[†], Kang, E.[†], **Moon, J.[†]**, Kim, H., Han, S., Choung, S., Kim, S., Yi, S. Y., Kang, E., Choi, C. H.* , Han, J. W.* , Lee, J.* (2025). **Quantity effect of heteroatom incorporation on the oxygen evolution mechanism in ruthenium oxide.** *Chem*, 11(5), 102367.
- [10] Choung, S.[†], Park, W.[†], **Moon, J.[†]**, Han, J. W.* (2024). **Rise of machine learning potentials in heterogeneous catalysis: Developments, applications, and prospects.** *Chemical Engineering Journal*, 494, 152757.

- [11] Choung, S.[†], Yang, H.[†], **Moon, J.**, Park, W., June, H., Lim, C., Han, J. W.* (2024). Theoretical tuning of local coordination environment of metal-nitrogen doped carbon catalysts for selective chlorine-evolution reaction. *Catalysis Today*, 425, 114358.
- [12] Choung, S., Kim, Y., **Moon, J.**, Roh, J., Hwang, J., Han, J. W.* (2023). Unveiling the catalyst deactivation mechanism in the non-oxidative dehydrogenation of light alkanes on Rh(111): Density functional theory and kinetic Monte Carlo study. *Catalysis Today*, 411-412, 113819.

CONFERENCE PRESENTATIONS

- [1] **Moon, J.** (2026). CatBench: Evaluating Accuracy, Robustness, and Efficiency of Machine Learning Interatomic Potentials for Adsorption Energy Predictions (Oral Presentation). *American Chemical Society (ACS)*, Chicago, United States.
- [2] **Moon, J.** (2026). Size-Dependent Surface States of Sub-Nanometer Ru Clusters Modulate Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution (Poster). *American Chemical Society (ACS)*, Chicago, United States.
- [3] **Moon, J.** (2026). Size-Dependent Surface States of Sub-Nanometer Ru Clusters Modulate Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution (Poster, Sci-Mix). *American Chemical Society (ACS)*, Chicago, United States.
- [4] **Moon, J.** (2026). Synergistic Effect of Oxygen Evolution Mechanism and Interfacial Water Dynamics Revealed by MLIP Distillation (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.
- [5] **Moon, J.** (2026). Machine Learning-Accelerated Large-Scale Analysis of Interfacial Water Structure on Size-Controlled Ru Clusters for Alkaline Hydrogen Evolution (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.
- [6] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *The 8th International Conference on Advanced Electromaterials (ICAE 2025)*, Jeju, South Korea.
- [7] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *Korean Institute of Metals and Materials (KIM)*, Gwangju, South Korea.
- [8] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [9] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Graph Neural Network (Oral Presentation). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [10] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Fine-Tuned Machine Learning Interatomic Potential (Oral Presentation). *Korean Institute of Chemical Engineers (KICHE)*, Jeju, South Korea.
- [11] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.
- [12] **Moon, J.** (2025). CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.
- [13] **Moon, J.** (2024). Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, PyeongChang, South Korea.

- [14] Moon, J. (2024). Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential (Oral Presentation). 23rd Korea Japan Students' Symposium, Seoul, South Korea.
- [15] Moon, J. (2024). Selective Control of Oxygen Evolution Reaction Mechanisms by Adjusting Mn Doping Concentrations in RuO₂ (Oral Presentation). The Korean Ceramic Society (KCerS), Busan, South Korea.

SKILLS

- **Programming & Frameworks:** Python (Advanced), PyTorch, Bash scripting (Intermediate)
- **Machine Learning:** MLIP Training & Benchmarking (20+ architectures), Dataset Generation, Bayesian Optimization
 - *MLIP Fine-tuning:* Generated 100,000+ in-house DFT calculations, fine-tuned GemNet-OC reducing adsorption energy MAE from 0.48 eV to 0.05 eV (evaluated by CatBench)
 - *Large-scale Molecular Dynamics:* MLIP-accelerated MD simulations (100+ ns) for metal dispersion and structural evolution analysis
 - *High-throughput Screening:* MLIP-accelerated screening of alloy catalysts using bulk structures from Materials Project, OQMD, GNoMe, and OMat datasets
 - *Genetic Algorithm:* Ensemble-GA development with sub-population partitioning and mixing strategies for enhanced global optimization
- **Computational Chemistry:** Density Functional Theory (VASP), Machine Learning Interatomic Potentials (20+ models: UMA, EquiformerV2, GemNet-OC, MACE, etc.), Molecular Dynamics, Electronic Structure Analysis
- **High-Performance Computing:** Supercomputer Cluster Server Management, SLURM, PBS, Batch Job Optimization
- **Research Expertise:** Heterogeneous Catalysis, Single Atom Catalysis, Electrocatalysis (OER, HER, CER), Multiscale Simulation using MLIP, MLIP Benchmarking, ML Analysis of Experimental Characterization
- **Language:** Korean (Native), English (Fluent)

TEACHING

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|---|---------------------------------|
| • Seoul National University
TA - Computer Concepts and Practice | Fall 2026
Seoul, South Korea |
| • Seoul National University
TA - Computer Concepts and Practice | Fall 2025
Seoul, South Korea |
| • Seoul National University
TA - Computational Materials and Data Science | Fall 2024
Seoul, South Korea |
| • Seoul National University
TA - Physical Chemistry of Materials | Fall 2023
Seoul, South Korea |